

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We cover a new method used to 3D print human tissue without scaffolding, delve into Chandrayaan 2 and attempt to unriddle the fermi paradox.



Evolution of puppy eyes

Dogs have evolved special muscles above their eyes, which let them raise their inner eyebrows. This appearance allows them to evoke nurturing feelings from humans. This muscle is not present in the wild relatives of dogs, including wolves. Research also has shown that the dogs increase the use of the muscle significantly, when in the presence of humans. <https://digit.in/puppyeyes>

WHAT'S NEW

Spidey sense for drones

Drones and autonomous vehicles need to respond quickly to tricky or dangerous situations. Unlike the slow human and machine reactions, animals do have the ability to respond rapidly. Spiders, bats, and other creatures have special neurons called mechanoreceptors which are sensors for critical life-saving information. They are in the form of hair, cilia and feathers, which only kick in during life-threatening scenarios. They filter unnecessary data. On a spider's leg, there are sensitive mechanosensors tuned to detect the vibrations associated with prey or mates. If dust settles on the spider web, the sensors will not detect that information, as it is irrelevant.



Researchers are attaching sensors on the outer shell of autonomous vehicles that are tuned to detect obstacles. These sensors are inspired by the fine hairs on legs of the spiders. They can be tuned to detect particular forces, which are associated with objects that need to be avoided. <https://digit.in/spidercar>

Unmasking deepfakes via facial quirks

Many methods are available to produce fake videos of famous people. Lip-sync, face-swap and puppet-master techniques use an audio-video recording of one person, and the photo of another, well-known individual to produce deep fake videos. Lip-sync technique uses a base audio to manipulate the movement of lips of real videos of a person, to make it seem like they are in sync.



The face-swap technique adds a digital mask atop a video of another person speaking. The puppet-master technique manipulates both the lips and jaws, by tracking an audio track. The resulting "deepfake" videos can be abused. New

digital forensics techniques are being used to identify deepfake videos by finding out particular quirks of the speakers. For example, Obama usually purses his lips after saying "hi everybody". These facial quirks can be used to distinguish fake videos from real ones. It is a powerful tool to fight against misinformation. The technique, however, works best in official videos. <https://digit.in/unmskfk>



How we stay focused

Many times we need to focus on certain conversations or perform some tasks in noisy environments. Neuroscientists have discovered a circuit in human brains that filters out unwanted sensory inputs. The prefrontal cortex filters out stimuli to minimise distractions. <https://digit.in/brainfocus>



Mammals thrived alongside dinos

It is a myth that mammals could only evolve after the extinction of dinosaurs opened up ecological niches. Paleontologists identified three diversification events, where mammals evolved into new ecological niches. Two of which happened during the age of the dinosaurs. <https://digit.in/mamrad>



Snail inspires superglue

Adhesives are of two types. Weak glue, which is easily reversible, and strong glue, which isn't. Weak glue can be reused unlike strong glue. Researchers, inspired by the mucous of snails, combined dry and wet materials to create a strong glue, which is reversible and reusable. <https://digit.in/snailglue>